

CANGZHOU, China

WMO: 546160

Lat: 38.333N

Lon: 116.833E

Elev: 11

StdP: 101.19

Time Zone: 8.00 (E08)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.4	-7.7	-25.4	0.4	-2.2	-22.3	0.5	-2.0	9.2	-0.8	8.1	-1.2	1.7	340	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	34.2	23.2	33.0	23.5	31.9	23.4	27.4	31.0	26.6	30.0	25.7	29.1	3.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	29.8	25.6	20.9	28.9	24.8	19.8	28.0	87.4	31.2	83.4	30.1	80.0	29.2	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.3	6.2	-12.2	37.3	2.7	1.5	-14.2	38.3	-15.8	39.2	-17.3	40.0	-19.2	41.0		
			-13.4	28.7	2.1	1.1	-15.0	29.5	-16.2	30.1	-17.4	30.7	-18.9	31.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	-1.9	1.3	6.7	15.0	20.3	25.2	26.8	26.1	21.7	15.1	6.3	0.0
	DBStd	10.76	2.89	3.27	3.89	3.89	3.75	2.78	2.23	2.03	2.75	3.83	4.60	2.99
	HDD10.0	1175	368	245	116	6	0	0	0	0	0	6	127	308
	HDD18.3	2649	627	478	359	114	24	1	0	0	6	114	361	567
	CDD10.0	2496	0	1	15	154	319	456	521	499	351	163	16	0
	CDD18.3	927	0	0	0	13	84	207	263	241	107	12	0	0
	CDH23.3	8787	0	0	2	110	792	2083	2730	2302	696	72	0	0
	CDH26.7	3293	0	0	0	18	272	910	1107	822	155	9	0	0
Wind	WSAvg	2.8	2.1	2.6	3.2	3.7	3.5	3.1	2.8	2.3	2.6	2.6	2.5	2.1
Precipitation	PrecAvg	613	3	5	8	24	35	72	218	162	47	22	12	4
	PrecMax	1161	22	22	45	164	118	156	451	411	149	70	33	25
	PrecMin	246	0	0	0	0	2	7	44	9	3	3	0	0
	PrecStd	200	5	6	10	30	30	41	99	101	36	17	10	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	15.6	21.1	28.9	34.0	36.9	35.7	33.8	31.6	28.0	19.9	10.7
		MCWB	2.1	6.4	10.9	16.6	19.2	20.9	23.7	26.0	22.5	18.2	11.5	5.6
	2%	DB	6.1	11.2	18.1	26.2	31.6	34.4	34.2	32.6	29.8	25.3	17.7	8.0
		MCWB	0.6	4.4	9.1	15.1	19.1	21.3	25.0	24.9	21.1	16.7	9.8	3.1
	5%	DB	4.4	8.8	15.7	24.2	29.6	33.0	33.0	31.7	28.4	23.3	15.2	6.2
		MCWB	-0.4	2.8	8.1	14.5	18.5	21.3	25.3	24.3	20.3	15.2	8.9	2.0
	10%	DB	2.8	6.8	13.7	22.2	27.8	31.5	31.7	30.8	27.2	21.4	12.9	4.7
		MCWB	-1.2	1.5	6.8	13.1	17.8	21.0	25.1	23.9	19.5	14.3	7.7	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	7.2	12.5	18.9	22.8	25.7	29.2	27.7	25.2	19.3	12.5	5.9
		MCDB	7.5	14.2	18.4	25.6	29.0	30.7	33.1	32.0	28.3	25.5	17.2	10.4
	2%	WB	1.5	4.7	10.4	16.7	21.1	24.5	27.9	26.8	24.0	17.9	11.2	4.1
		MCDB	4.9	10.7	16.4	24.4	28.4	30.5	31.5	30.6	27.1	23.2	15.3	6.8
	5%	WB	0.2	3.3	8.7	15.6	20.0	23.2	27.2	26.1	22.4	16.7	9.9	2.8
		MCDB	2.9	7.5	14.7	22.4	27.1	29.0	30.6	29.5	26.2	21.3	13.8	5.2
	10%	WB	-0.7	2.1	7.3	14.3	19.0	22.4	26.4	25.5	21.3	15.5	8.5	1.6
		MCDB	2.1	5.6	12.8	20.4	25.2	28.1	29.7	28.8	25.3	19.9	12.4	4.2
Mean Daily Temperature Range		MDBR	8.0	9.0	9.3	10.7	10.9	10.3	7.9	7.7	9.3	9.7	8.7	7.5
	5% DB	MCDBR	10.3	12.2	12.6	13.0	13.9	12.9	9.8	8.9	10.6	11.4	11.8	9.9
		MCWBR	6.3	7.4	6.7	5.8	5.5	4.5	3.7	3.1	4.1	5.0	6.3	6.2
	5% WB	MCDBR	8.6	10.8	11.4	12.2	11.7	10.6	7.8	7.0	8.5	9.8	10.1	8.6
		MCWBR	5.9	6.8	6.5	6.4	5.9	4.9	4.1	3.2	4.1	4.9	6.3	6.0
Clear-Sky Solar Irradiance	taub	0.492	0.565	0.581	0.631	0.672	0.809	0.798	0.758	0.684	0.663	0.555	0.500	
	taud	1.793	1.680	1.664	1.609	1.573	1.397	1.458	1.510	1.585	1.579	1.730	1.821	
	Ebn at Noon	620	630	680	679	663	577	578	585	594	541	554	576	
	Edh at Noon	161	205	229	256	270	322	301	279	245	224	171	147	
All-Sky Solar Radiation	RadAvg	2.35	3.03	4.44	5.30	5.92	5.35	4.79	4.58	4.12	3.19	2.43	2.12	
	RadStd	0.25	0.40	0.39	0.30	0.42	0.48	0.33	0.34	0.36	0.32	0.36	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page